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Short tandem repeats (STRs) are short sequences of DNA, usually 2 to 7 base pairs. STRs are repeated a number of times, one after another. For example, the STR D5S818 is made up of AGAT repeated 7 to 16 times.

STRs are found throughout the whole genome.

The repeated sequences in STRs are common to all humans. Due to variation in the number of repeats, STRs can be used in genetic fingerprinting.

0 9**1**

Describe how STRs could be removed from a sample of DNA.

[2 marks]

0 9**2**

Genetic fingerprinting using STRs requires amplification of the STRs using the polymerase chain reaction (PCR). The short base sequences either side of a specific STR are known.

Explain the importance of knowing these base sequences in PCR.

[2 marks]

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A single STR molecule consisting of a 12 **base pair** sequence of AGAT was amplified 50 times using PCR.

Calculate the total number of base pairs in all the STR molecules after 50 cycles of PCR.

Show your working.

[2 marks]

Answer _____

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During genetic fingerprinting, the different STRs are separated by gel electrophoresis.

Give **two** features of STRs which enable them to be separated by gel electrophoresis.

[2 marks]

1 _____

2 _____

Turn over for the next question

